MODULE 19 · SECTION 3 - TECHNIQUES TO IMPROVE GEN AI MODEL OUTPUT

Where models fail, and the fix for each

Six limitations the exam guide names, how to spot each one in a question, and the matching fix.

SECTION 3 of 4 EXAM WEIGHT ~20% MODULE 19 of 33

admin@sthithapragna.com

The exam describes a **symptom**. Your job is the diagnosis, and then the treatment.

admin@sthithapragna.com

Six limitations of foundation models

LIMITATION	WHAT GOES WRONG
Hallucination	Confident answers that are false
Knowledge cutoff	Knows nothing after its training date
Data dependency	Only as good as the data it learned from
Bias	Skewed patterns copied from the data
Fairness	Outcomes that treat groups unequally
Edge cases	Unusual inputs it rarely or never saw

ON THE EXAM

Learn these six as a set. Questions describe one and offer fixes for the others as distractors.

Hallucination: fluent, confident and wrong

- **The model predicts likely words;** it does not look facts up.
- **So it can invent** a policy, a figure, a citation or a product.
- **It sounds just as confident** when it is wrong.
- **Bigger models** hallucinate less, but never zero.
- **Wrenfield quirk 7:** a pilot assistant said returns last 60 days. It is 30.

THE SIGNAL

A plausible but invented fact, stated with confidence: hallucination. The fix is grounding, not a bigger model.

Knowledge cutoff: a letter from the past

- **Training stops** on a date; the model knows nothing after it.
- **New products, prices, news** and rules are missing.
- **It may not know** that it does not know.
- **Retraining** to catch up is slow and costly.
- **Wrenfield quirk 15:** the model has never heard of the autumn 2026 range.

THE SIGNAL

Asks about something recent, and the model is out of date: knowledge cutoff. Fix with grounding in fresh sources.

Data dependency: only as good as its data

- **A model learns** from the data it was trained on, and nothing else.
- **Gaps in that data** become gaps in its answers.
- **Errors in that data** become errors in its answers.
- **Your private data** was never in it: your prices, stock or policies.
- **The six marks of quality** from module 4 still apply.

THE SIGNAL

Answers are poor because the data was thin, wrong or missing: data dependency. Fix the data, or ground in better data.

Bias or fairness?

BIAS

A skew in the data or model

- Patterns copied from history
- Over or under-represented groups
- Often invisible in a demo
- The cause

FAIRNESS

A harm in the outcome

- Decisions that disadvantage a group
- Measured on real results
- Can break laws and trust
- The effect

THE TIEBREAKER

Bias is usually the cause; an unfair outcome is the effect. Fix with representative data, fairness testing, and people reviewing decisions.

Edge cases: the inputs nobody planned for

- **Rare inputs** the model seldom or never saw in training.
- **At Wrenfield:** a complaint in Welsh, a discontinued range, a part refund.
- **Behaviour becomes unpredictable** exactly where it matters most.
- **A great demo** can hide them, because demos use common cases.
- **Fix:** test deliberately with unusual inputs, and hand off to a person.

THE SIGNAL

Works on common requests but fails on rare ones: edge cases. Test for them, and route them to a human.

Five fixes, cheapest first

FIX	WHAT IT DOES
Prompt engineering	Clearer instructions, examples and constraints
Grounding	Ties answers to trusted, verifiable sources
RAG	Retrieves relevant passages, then generates
Fine-tuning	Teaches a style, format or task with examples
Human in the loop	People review, approve or correct outputs

REMEMBER

Grounding is the goal; RAG is the most common way to reach it. Both fix facts. Fine-tuning fixes behaviour.

The fix for each limitation

LIMITATION	FIRST-CHOICE FIX
Hallucination	Grounding or RAG; people check high stakes
Knowledge cutoff	Ground in fresh sources, such as Google Search
Data dependency	Better data, or ground in your own data
Bias	Representative data and evaluation
Fairness	Fairness testing; people make the decision
Edge cases	Test unusual inputs; hand off to a person

THE TIEBREAKER

A bigger model, or a lower temperature, is almost never the first-choice fix. Look for the option that addresses the cause.

RAG or fine-tuning?

RAG AND GROUNDING

Fixes what it knows

- Facts, policies, prices, stock
- Changes often
- Answers can cite a source
- Update the documents, not the model

FINE-TUNING

Fixes how it behaves

- Tone, format, a specialised task
- Stays stable over time
- Needs good labelled examples
- Retrain to change it

THE TIEBREAKER

Facts that change: RAG. A style or format prompts cannot hold: fine-tuning. An open-book exam versus cookery school.

Human in the loop: where to put the people

- **Before action** when stakes are high: refunds, credit, hiring.
- **On low confidence:** the system hands off when it is unsure.
- **By sampling:** reviewers check a share of routine outputs.
- **Through feedback:** corrections improve prompts, data and tests.
- **Not on everything:** that removes the benefit of automation.

THE TAKEAWAY

High-stakes, regulated or irreversible decisions keep a person in the loop. Routine outputs get sampled, not checked one by one.

Wrenfield's failures, diagnosed and treated

SYMPTOM	DIAGNOSIS AND FIX
"Returns last 60 days" (it is 30)	Hallucination: ground in the current policy
Knows nothing of the autumn range	Cutoff: RAG over the live catalogue
Hiring pilot favours some postcodes	Bias and fairness: audit; people decide
Welsh-language complaints go wrong	Edge case: test, then hand off
Copy drifts from house style	Behaviour: few-shot prompts, then fine-tune

THE TAKEAWAY

Five symptoms, five different treatments. No single fix solves them all, which is why options that promise one cure are suspect.

> Wrenfield's pilot assistant told a customer the returns window is 60 days. The real policy, 30 days, sits in the staff handbook.
>
> The team wants answers that match the current policy, with a source customers can check.

What is the best fix?

A Switch to the largest available model

B Ground answers in the current returns policy, with citations

C Raise the temperature for more varied answers

D Fine-tune on last year's chat transcripts

◆ PAUSE AND DECIDE

What is the best fix?

A Switch to the largest available model

Hallucinates less, but still guesses.

B Ground answers in the current returns policy, with citations

Answers tied to the real policy, with a source.

C Raise the temperature for more varied answers

More variety means more invention, not less.

D Fine-tune on last year's chat transcripts

Teaches style, and bakes in last year's errors.

THE SIGNAL

Confident but invented facts: ground in trusted sources.

> Wrenfield's product assistant knows nothing about the autumn 2026 range.
>
> The range changes every season, and stock and prices change daily.

What is the best fix?

A Fine-tune the model every season

B Tell the model to answer confidently

C Train a new foundation model

D RAG over the live product catalogue

◆ PAUSE AND DECIDE

What is the best fix?

A Fine-tune the model every season

Slow, costly, and stale within days.

B Tell the model to answer confidently

Confidence without knowledge invents answers.

C Train a new foundation model

Vast cost, and out of date on launch day.

D RAG over the live product catalogue

Fresh facts at answer time, updated as the catalogue changes.

THE SIGNAL

Facts after the cutoff, changing often: RAG over fresh data.

> Asked about delivery rules, Wrenfield's assistant cites "the Furniture Delivery Standards Act 2019", section 4.
>
> No such act exists, but the answer reads perfectly.

Which limitation does this show?

A Hallucination

B Knowledge cutoff

C Bias

D An edge case

◆ PAUSE AND DECIDE

Which limitation does this show?

A Hallucination

An invented fact, stated with confidence.

B Knowledge cutoff

Cutoff means missing recent facts, not inventing laws.

C Bias

Bias is a skew across groups, not a made-up source.

D An edge case

A common question, not a rare input.

THE SIGNAL

A fluent, confident invention: hallucination.

> Wrenfield's copy assistant already receives every product fact in the prompt.
>
> But after weeks of prompt work, its descriptions still drift from the strict house style and fixed layout. The team has 5,000 approved examples.

What should they try next?

A RAG over the product catalogue

B Grounding with Google Search

C Fine-tune on the approved examples

D A person rewrites every description

◆ PAUSE AND DECIDE

What should they try next?

A RAG over the product catalogue
The facts are already supplied; style is the problem.

B Grounding with Google Search
Brings in web facts, not house style.

C Fine-tune on the approved examples
Teaches a stable style and format from examples.

D A person rewrites every description
Works, but removes the benefit at this volume.

THE SIGNAL

Behaviour that prompts cannot hold, with good examples: fine-tuning.

> Wrenfield's hiring pilot ranked applicants from some postcodes consistently lower, though their experience matched others.
>
> The pilot learned from five years of past hiring decisions.

Which diagnosis and response fit best?

A Hallucination: ground the ranking in the CVs

B Knowledge cutoff: retrain on this year's data

C Edge case: add more test applicants

D Bias causing unfair outcomes: audit the data and keep people deciding

◆ PAUSE AND DECIDE

Which diagnosis and response fit best?

A Hallucination: ground the ranking in the CVs

Nothing invented; the pattern was learned.

B Knowledge cutoff: retrain on this year's data

Recency is not the problem; history is.

C Edge case: add more test applicants

A whole group is affected, not rare inputs.

D Bias causing unfair outcomes: audit the data and keep people deciding

Learned skew from history, harming a group.

THE TAKEAWAY

A learned skew that harms a group: bias and fairness.

> Wrenfield's returns agent handles common requests well.
>
> But it mishandles rare ones: messages in Welsh, items from ranges discontinued years ago, and part refunds.

What is the best approach?

A Switch the agent off

B Test deliberately with unusual cases, and hand off to a person when unsure

C Use a bigger model

D Raise the temperature

◆ PAUSE AND DECIDE

What is the best approach?

A Switch the agent off

Loses the benefit on the common cases.

B Test deliberately with unusual cases, and hand off to a person when unsure

Finds the gaps, and keeps people for the rare ones.

C Use a bigger model

May help a little, but leaves rare cases untested.

D Raise the temperature

Adds randomness; no help with rare inputs.

THE SIGNAL

Fine on common cases, wrong on rare ones: edge cases.

Module 19 in five lines

- Six limitations: hallucination, cutoff, data dependency, bias, fairness, edge cases.
- Diagnose the symptom first; then choose the fix that treats the cause.
- Grounding and RAG fix facts; fine-tuning fixes behaviour.
- Bias is the cause; unfair outcomes are the effect. People decide high-stakes cases.
- A bigger model or lower temperature is rarely the first fix.